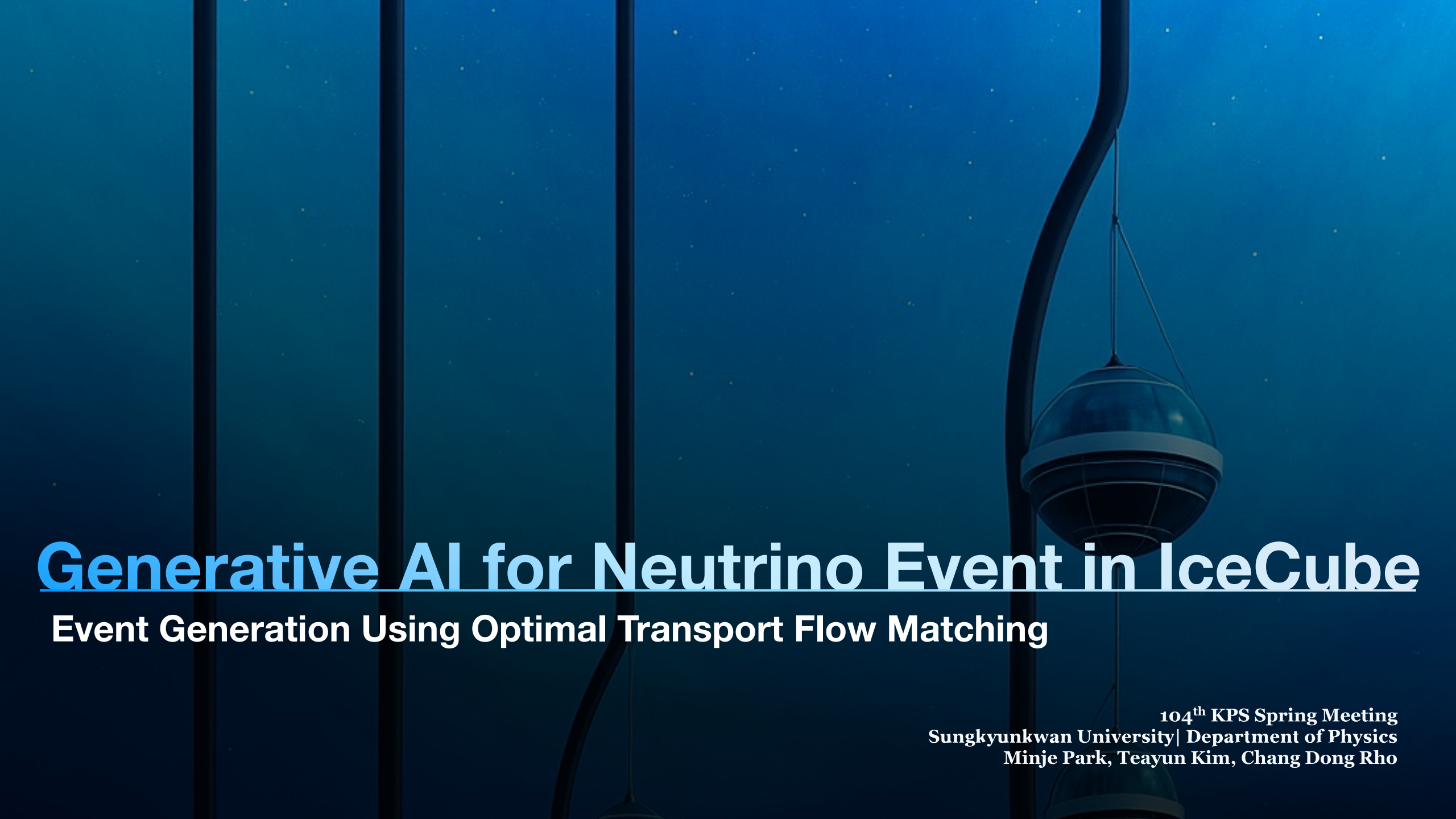


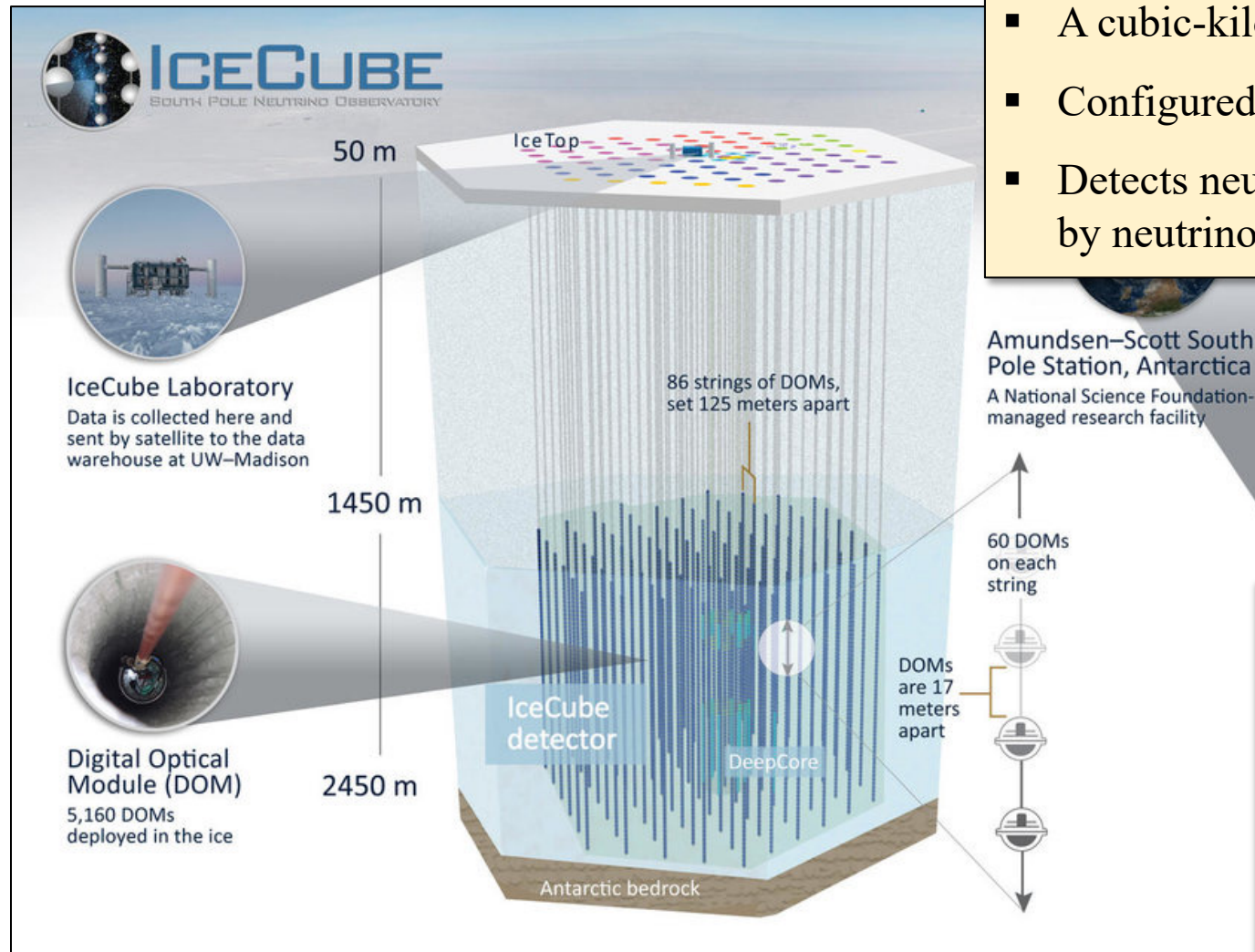


Generative AI for Neutrino Event in IceCube

Event Generation Using Optimal Transport Flow Matching

104th KPS Spring Meeting
Sungkyunkwan University | Department of Physics
Minje Park, Teayun Kim, Chang Dong Rho

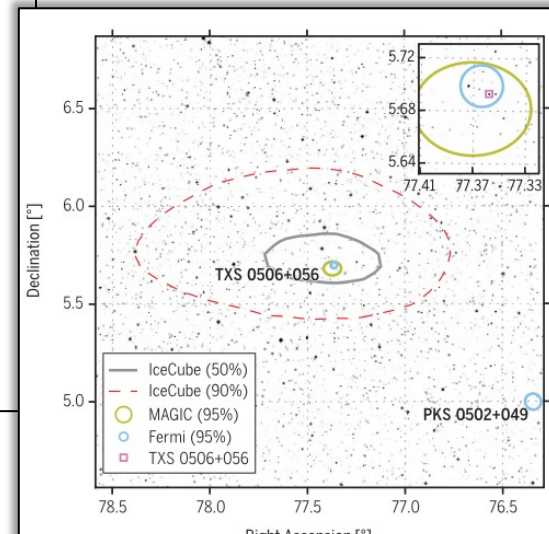
About IceCube



- A cubic-kilometer neutrino detector at the South Pole
- Configured with 5,160 Digital Optical Modules (DOM)
- Detects neutrino via Cherenkov radiation from particles produced by neutrino interaction

Major Achievements

- 2013: Discovery of astrophysical neutrinos
- 2018: First identified source — TXS 0506+056
- 2023: First neutrino map of the Milky Way

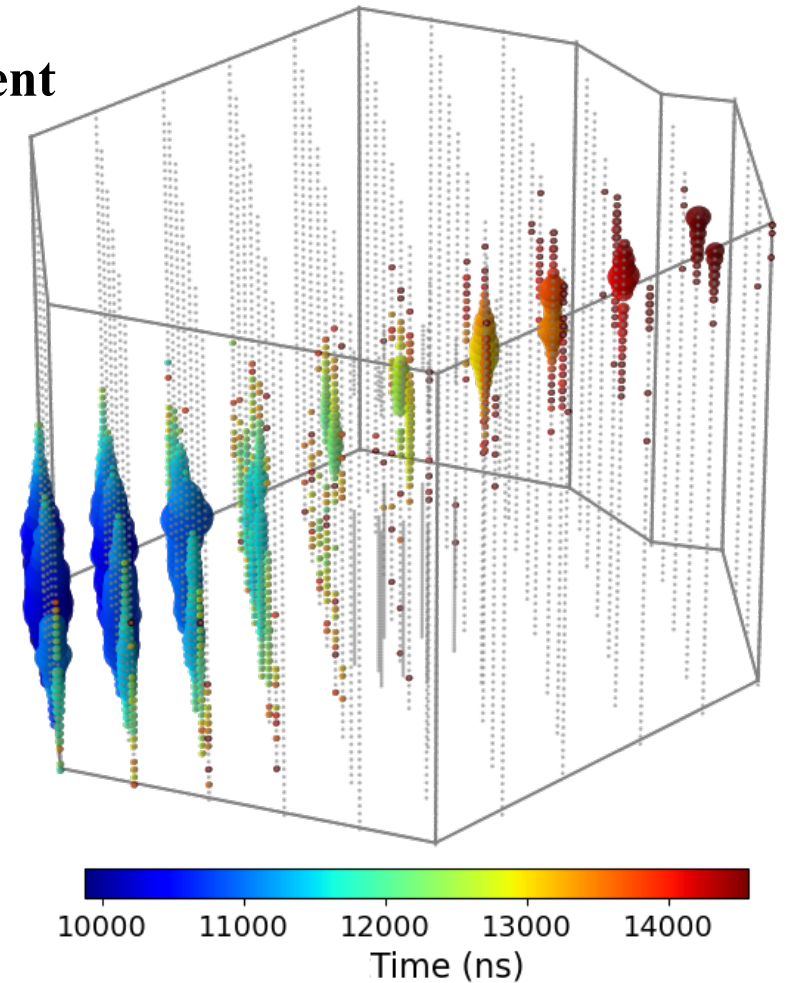
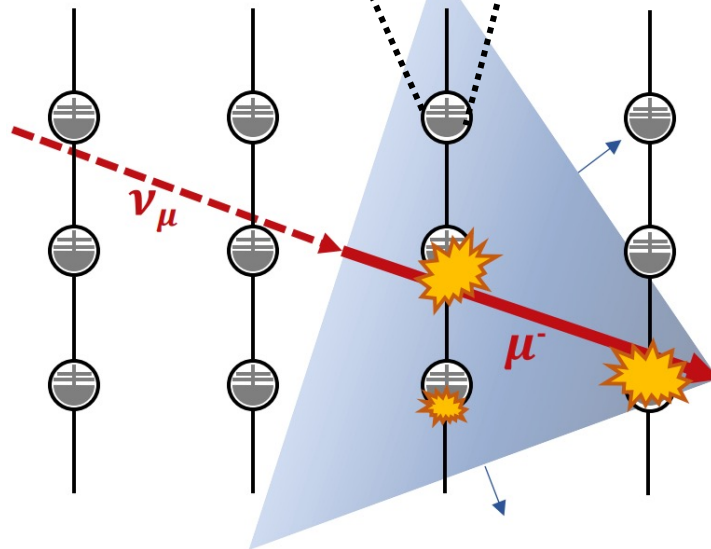
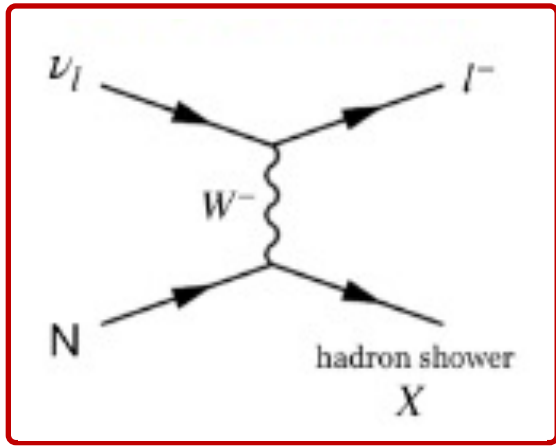
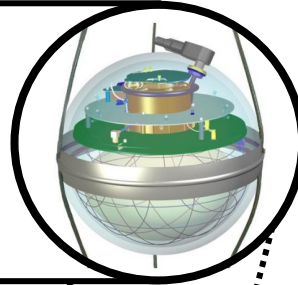


Track Signature

When ν_μ undergoes CC interaction near Icecube, the resulting μ passes through the detector, leaving a **track signatured event**

Each DOM records

- NPE : number of detected photoelectrons
- Hit Time : arrival time of earliest photon



The size of each DOM indicate the number of detected photons

Reconstruction Methods & Limitation

- Track events have an angular resolution of $< 1^\circ$ $\rightarrow$ Main channel for neutrino's source studies
- There are two main approaches to track event reconstruction in IceCube

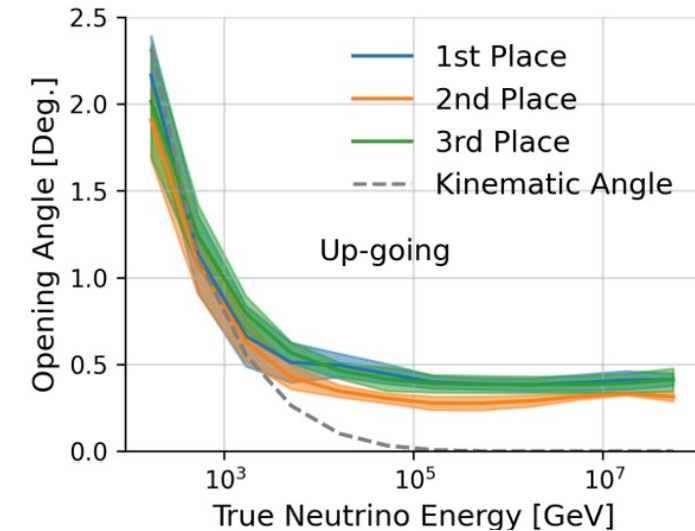
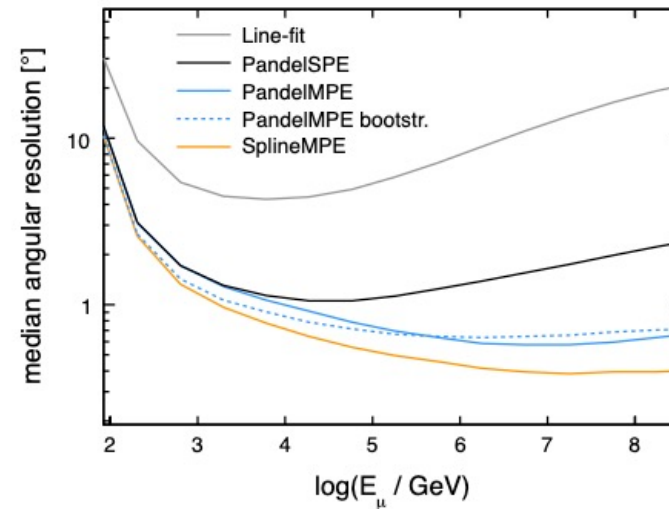
1. Spline Reconstruction [Left]

Maximum likelihood method using photon arrival time PDFs derived from simulations

2. Neural Network [Right]

Regression models that map observed events to track parameters

- However, both methods show a gap to the theoretical lower bound, with performance degradation around 1 TeV.
- We believe this gap is related to the difficulty of capturing the stochastic nature of high-energy events.
- Generative AI is known to learn the underlying structure of data more fundamentally.



Generative AI based-framework that can both:

- 1.Generate track events**
- 2.Reconstruct neutrino parameters**

Flow Matching & Generative AI

In general, Generative AI refers to models that can generate data such as images or videos
The objective of Generative AI is to transport known distribution p_0 to data distribution (q)

Flow Matching is a subset of Generative AI

learns a flow, ψ_t , that transport $p_0 \rightarrow q$

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x$$

$$\mathcal{L}_{\text{fm}} = \mathbb{E}_{x_t, t} [u_t^{\text{model}}(x_t) - u_t^{\text{target}}(x_t)]$$

$$\begin{aligned} \text{Optimal transport} \\ x_t &= tx_1 + (1-t)x_0 \\ \nabla \mathcal{L}_{\text{fm}} &= \nabla \mathcal{L}_{\text{OT-fm}} \end{aligned}$$

$$\mathcal{L}_{\text{OT-fm}} = \mathbb{E}_{x_1, x_0, t} [u_t^{\text{model}}(tx_1 + (1-t)x_0) - (x_1 - x_0)]$$

After training, we can generate data by follows:

```
Initialize  $x_0 \sim p_0$ ,
for  $t = 0 \rightarrow 1$  do
     $x_{t+\Delta t} = x_t + \Delta t \cdot u_t^{\text{model}}(x_t)$ 
Return  $x_1$ 
```

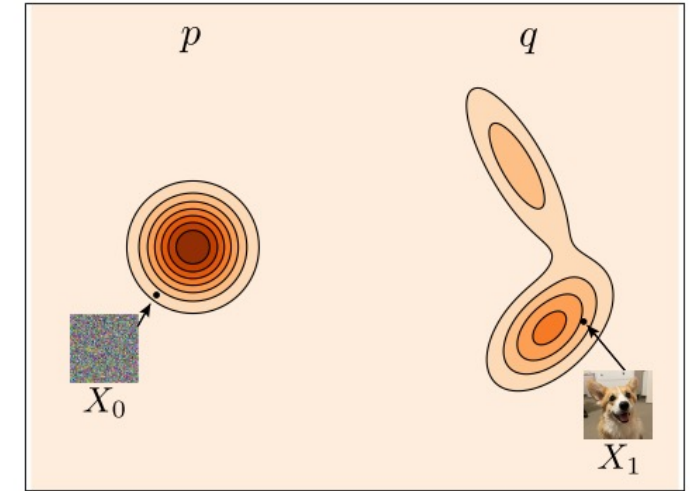
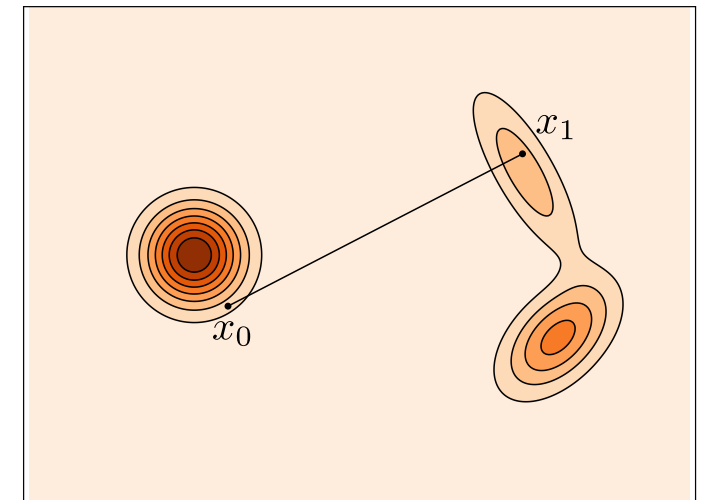


Figure of source distribution p_0 and data distribution q



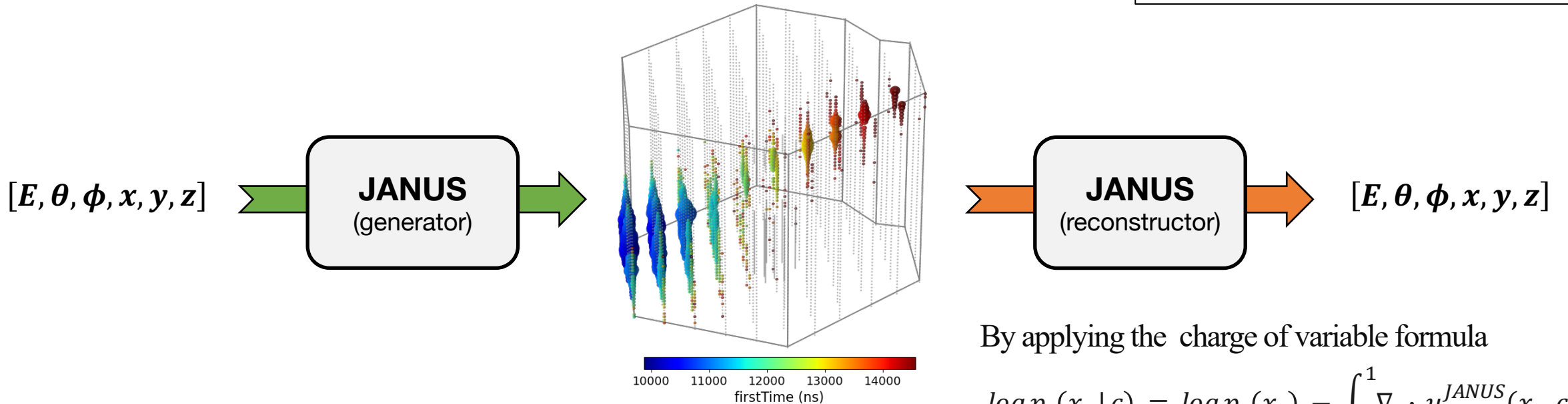
Optimal transport between p_0 to q

JANUS : Joint generative AI for Neutrino reconstruction and Simulation

- JANUS is trained to generate track events conditioned on neutrino parameters $c = [E, \theta, \phi, x, y, z]$

- After training, JANUS enables likelihood estimation without additional training

E : neutrino energy
 θ, ϕ : incoming zenith, azimuth angle
 x, y, z : track starting point



At this presentation, we focus on the generator component of JANUS

By applying the charge of variable formula

$$\log p_1(x_1 | c) = \log p_0(x_0) - \int_0^1 \nabla \cdot u_t^{JANUS}(x_t, c) dt$$

$$\arg \max \{ \log p_1(x_1 | c) \}$$

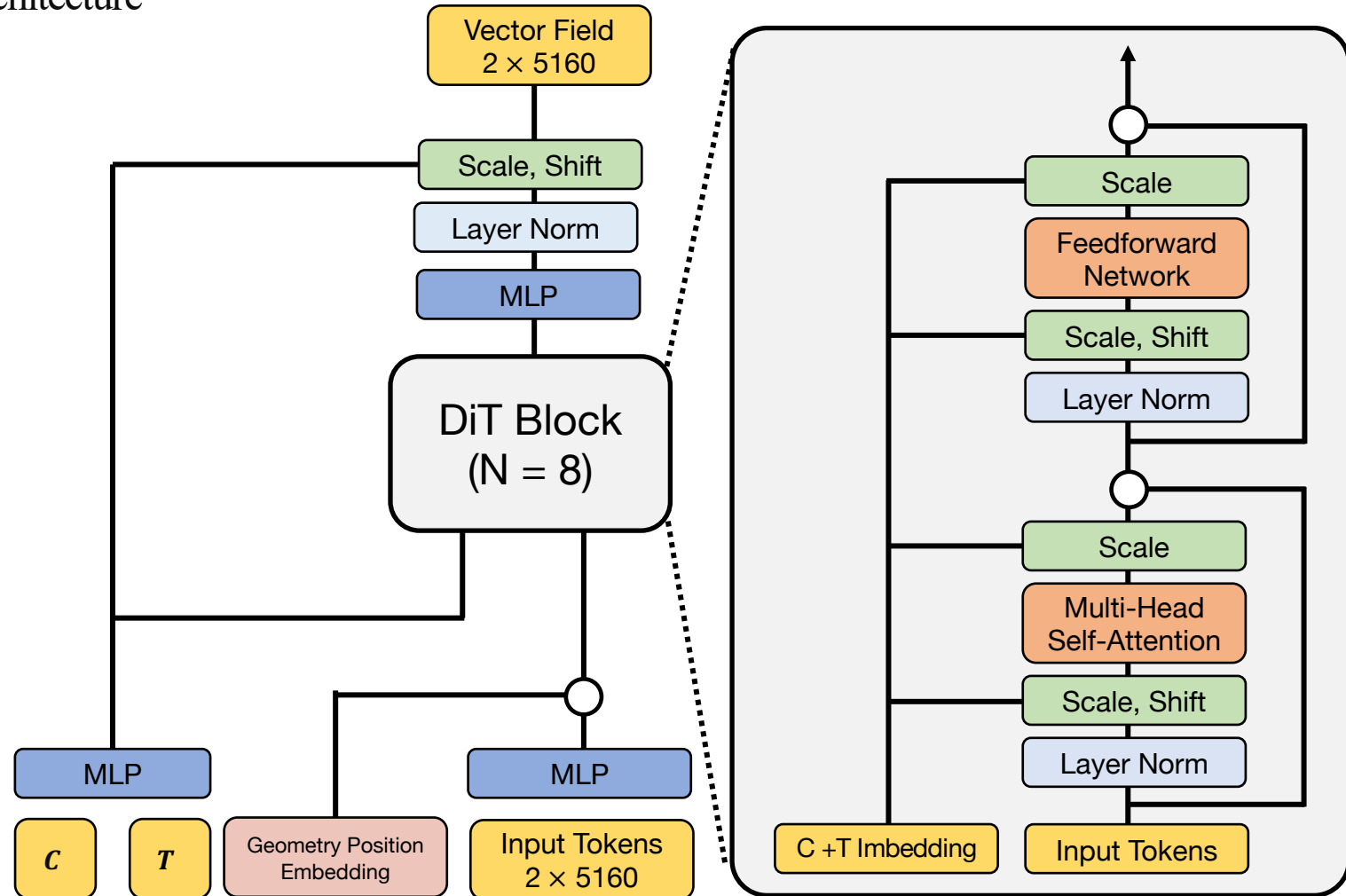
JANUS Architecture & Dataset

JANUS mainly follows DiT with AdaLN-Zero Architecture

Used detector geometry as position embedding

- Input : $tx_1 + (1 - t)x_0$
 - Output : $x_1 - x_0$ (Vector Field)
 - Condition(C) : $[E, \theta, \phi, x, y, z]$
 - Time Step(T) : $0 \sim 1$
-
- Icecube simulation dataset
 - Number of training data (events) : 178,056
 - Energy range : $1 \sim 100 \text{PeV}$
 - Up going track only

Trained on NVIDIA-A100 : 10hours

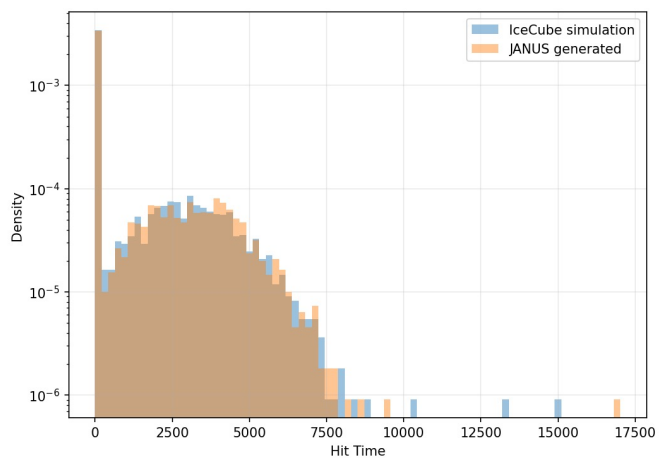
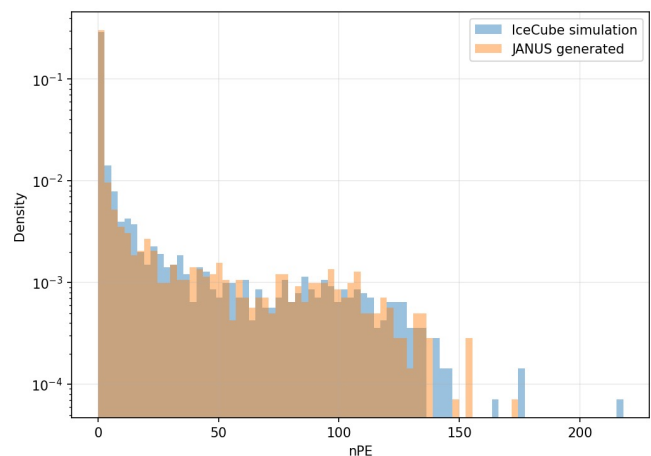


Generated Event 1

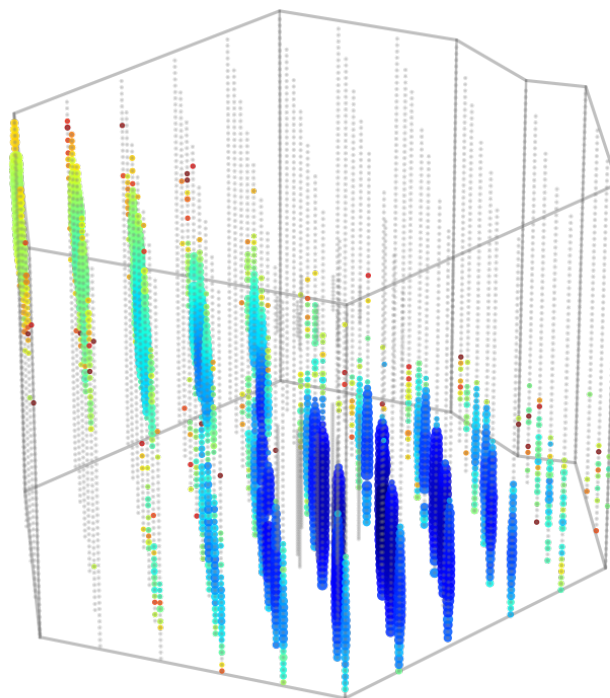
E : 42.224PeV

θ, ϕ : 133.946°, 8.633°

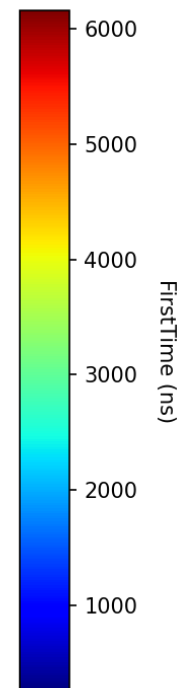
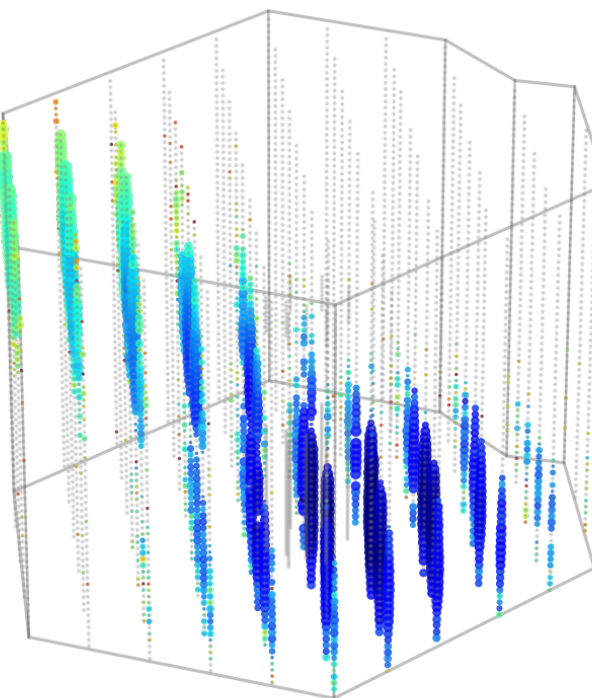
x, y, z : -248m, -111m, -284m



IceCube Simulation



JANUS Generated

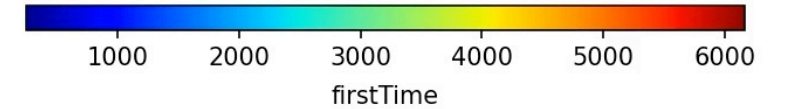
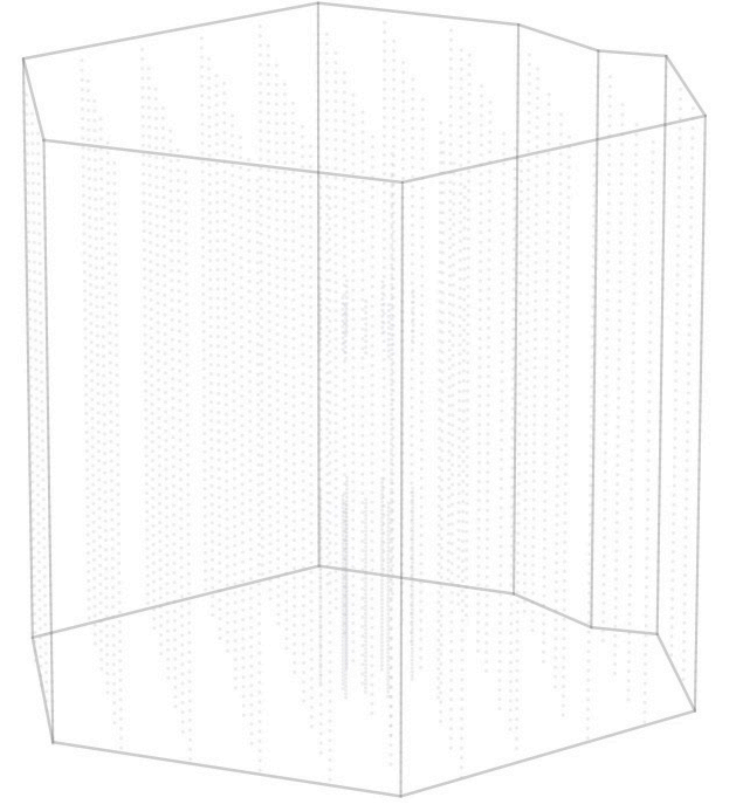
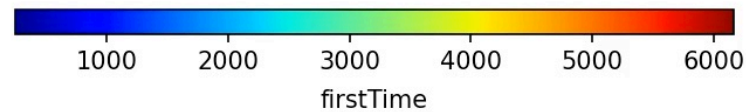
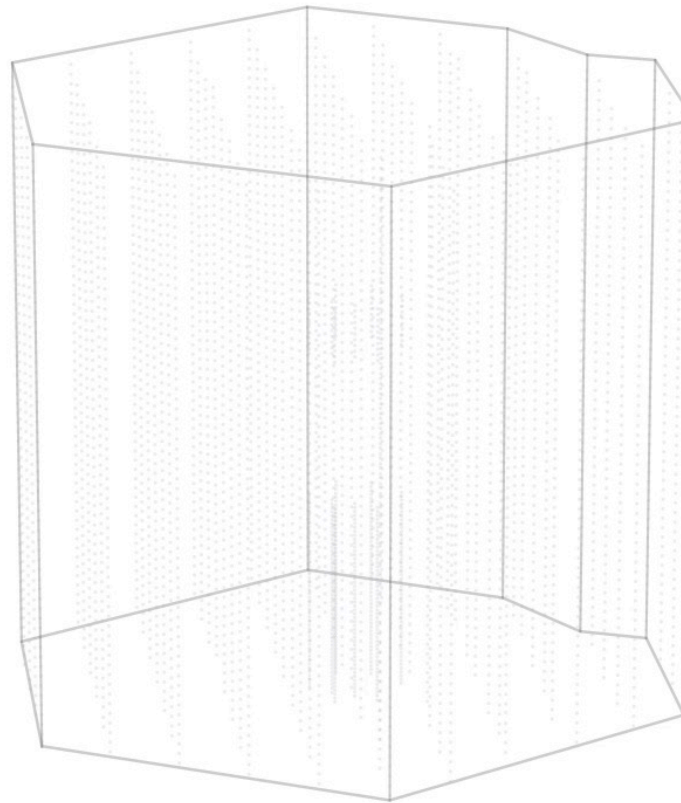


Generated Event 1 - Video

E : 42.224PeV

θ, ϕ : 133.946°, 8.633°

x, y, z : -248m, -111m, -284m

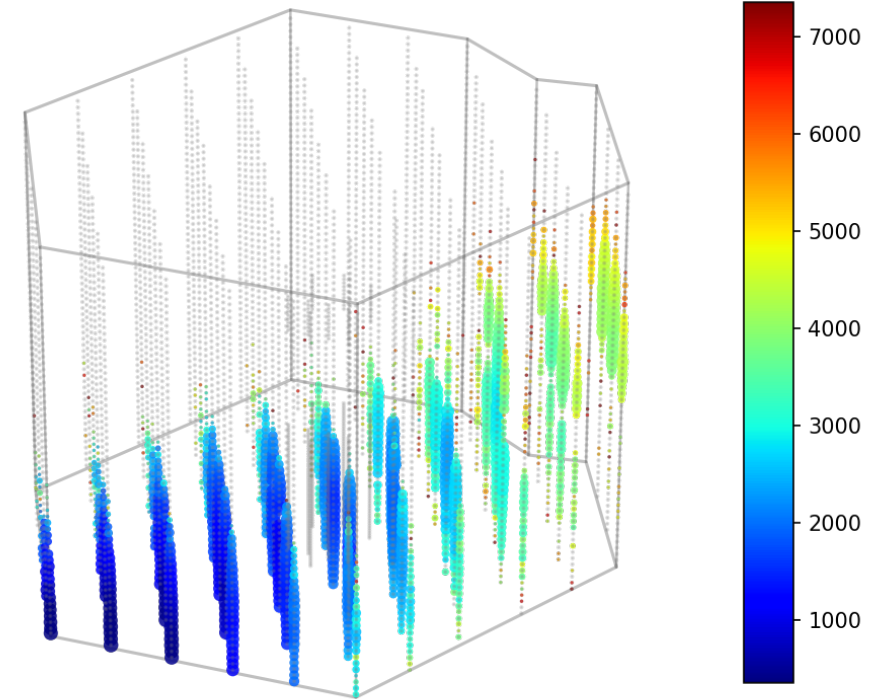
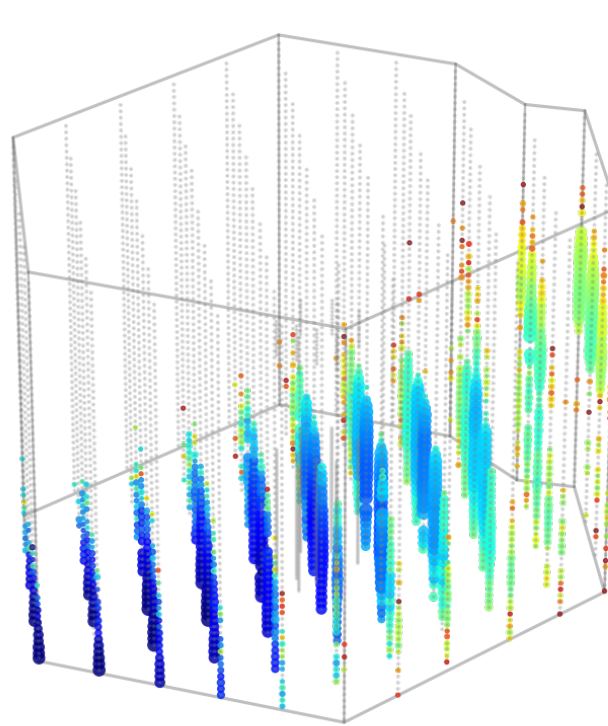
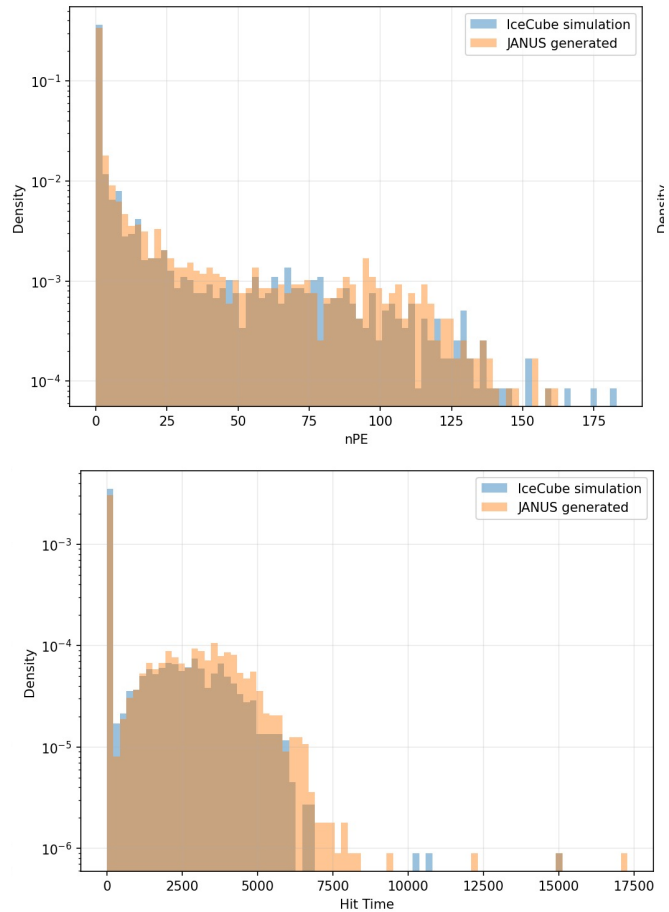


Generated Event 2

E : 19.960PeV
 θ, ϕ : 123.965, 233,903
 x, y, z : -215,-514,-499

IceCube Simulation

JANUS Generated



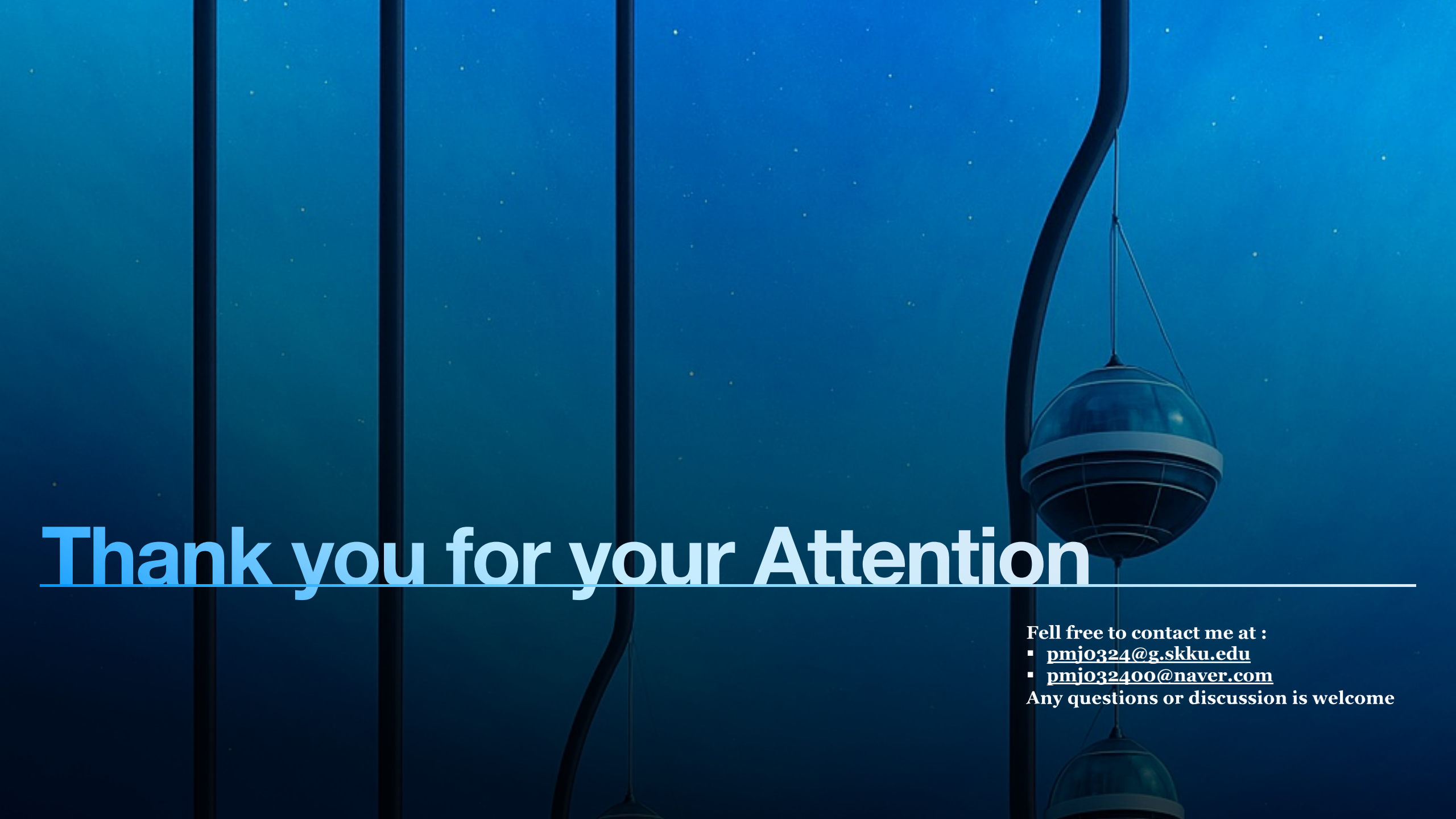
Summary & Future Work

Summary

- We trained JANUS using a optimal transport flow matching.
- To the best of our knowledge, this is the first application of generative AI to Icecube neutrino event
- JANUS captures the overall trends of track events

Future Work

- Design a summary statistic for the JANUS generated events
- Likelihood evaluation and parameter inference with JANUS

The background is a deep blue gradient with small white stars, suggesting a night sky. On the right side, there is a dark, curved structure with a spherical lantern hanging from it. The lantern has a grid-like pattern and a white band around its middle. Another similar lantern is visible below it.

Thank you for your Attention

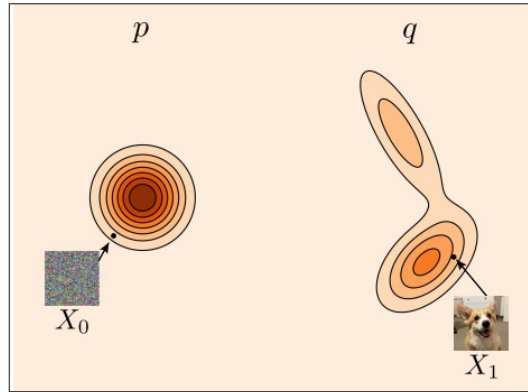
Fell free to contact me at :

- pmj0324@g.skku.edu
- pmj032400@naver.com

Any questions or discussion is welcome

Why Generative AI

- How can we infer from just training to sample the event



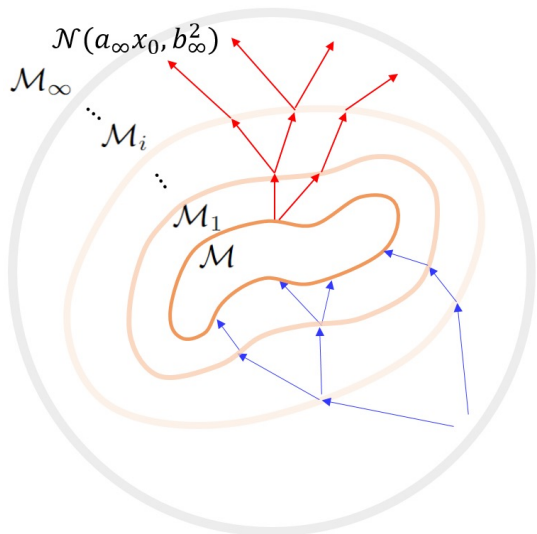
(a) Flow

- Deterministic transport (ODE): $\dot{x}_t = v_\theta(x_t, t)$
- $\log p_\theta(x_1 | c) = \log p_0(x_0) - \int \nabla_x v_\theta(x_t, t | c) dt$

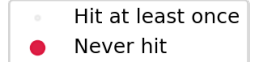
(b) Diffusion

- Stochastic noising / denoising (SDE / reverse process)
- $\log p_\theta(x_1 | c) \geq \mathbb{E}_{q(x_{0:1}|x_1)} [\log p_0(x_0) + \int_0^1 \log p_\theta(x_t | x_{t+dt}, c) dt - \int_0^1 \log q(x_t | x_{t+dt}) dt]$

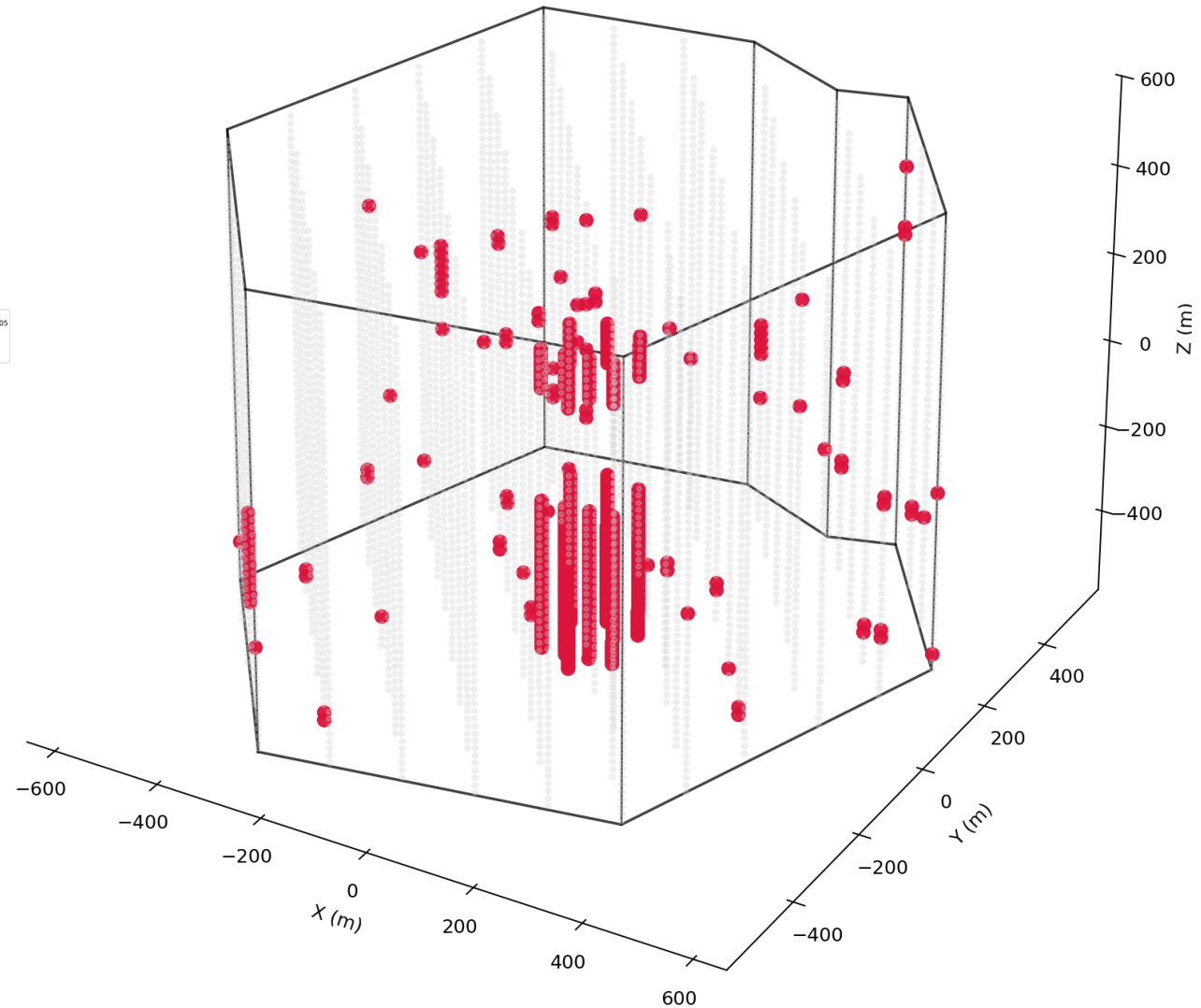
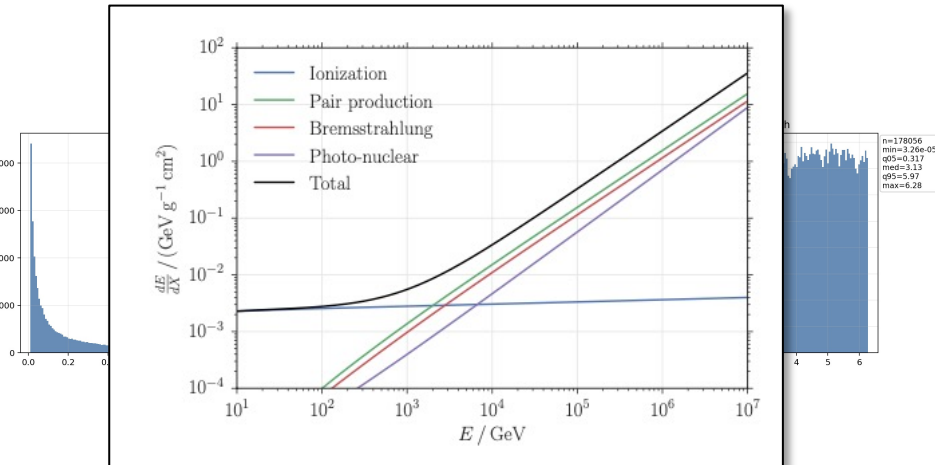
- Why should we use it rather than basic regression model



- Manifold hypothesis*: data lie on a low-dimensional manifold(q)
- Events from the same underlying muon cluster around the same manifold.
- A generative model learns q itself, enabling manifold-consistent, likelihood-based inference that is more robust to stochastic variability.



[1]



Flow Matching & Generative AI

In general, Generative AI is model that can generate data ; such as image or video

Objective of Generative AI is to model the data distribution (q)

Flow Matching is subset of Generative AI

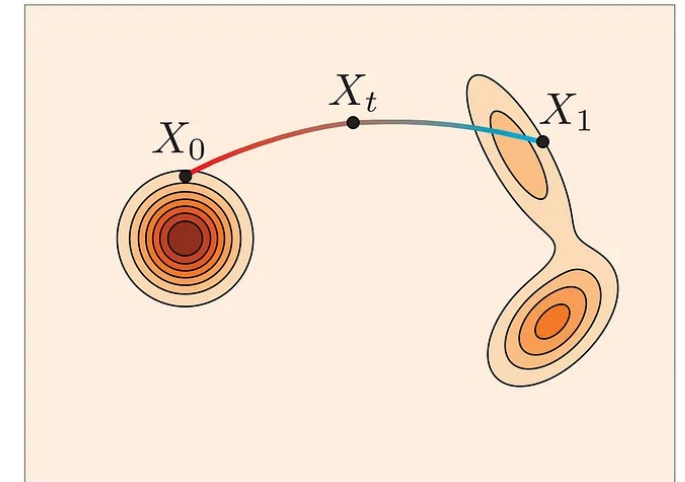
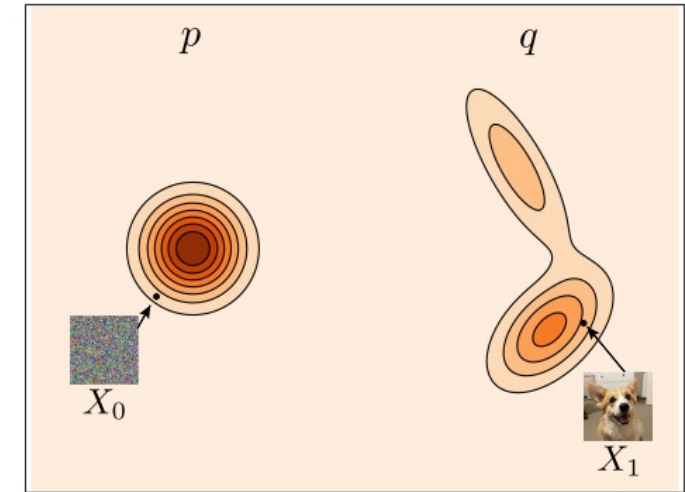
learns a flow, ψ_t , that the pushforward $p_0 \rightarrow q$

$$\frac{d}{dt}\psi_t(x) = u_t(\psi_t(x)), \quad \psi_0(x) = x$$

$$\mathcal{L}_{\text{fm}} = \mathbb{E}_{x_t, t} [u_t^{\text{model}}(x_t) - u_t^{\text{target}}(x_t)]$$

$$\mathcal{L}_{\text{cfm}} = \mathbb{E}_{x_1, x_0, t} [u_t^{\text{model}}(x_t) - \text{[redacted]}]$$

$$\nabla \mathcal{L}_{\text{fm}} = \nabla \mathcal{L}_{\text{cfm}}$$

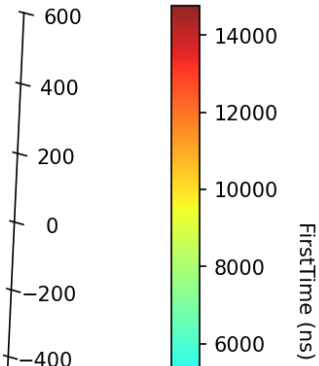
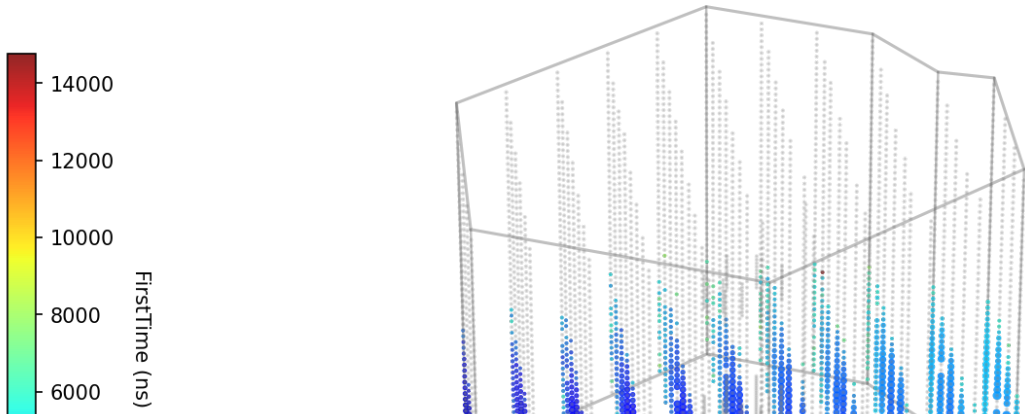


Never Hit PMT in 22644

[1]

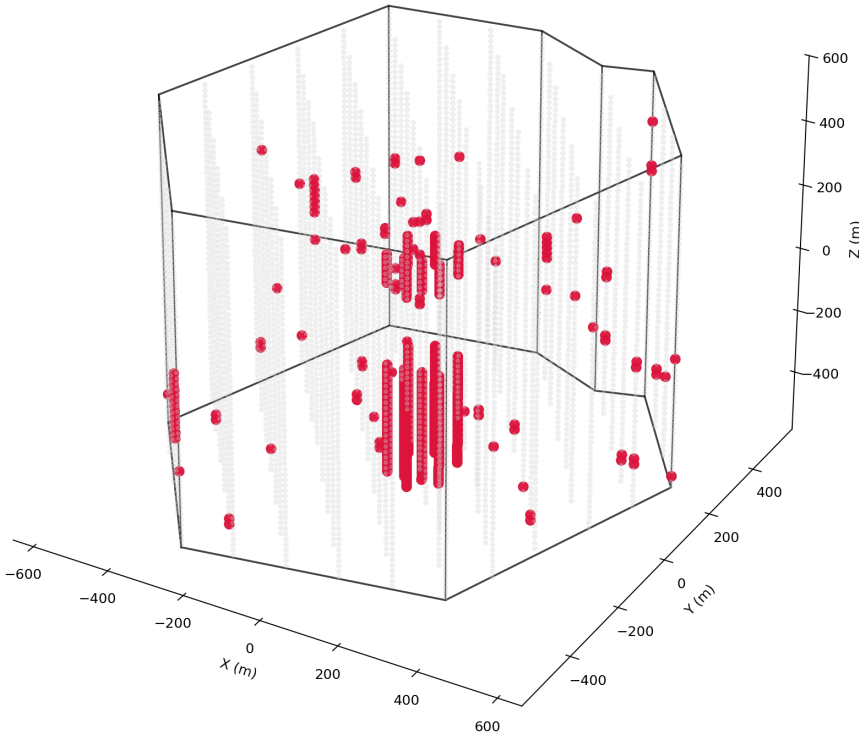
Rank 1 | ref=39276 | sample=1 | rk4/100 | cfg=1.0
V, ux=-0.771, uy=-0.605, X=-334.8, Y=-424.5, Z=-401.5

Generated event (denorm)



PMTs Never Hit
H5=22644_0921_time_shift.h5 | events=178,056 | PMTs=5,160 | mode=npe | never=584

Hit at least once
Never hit

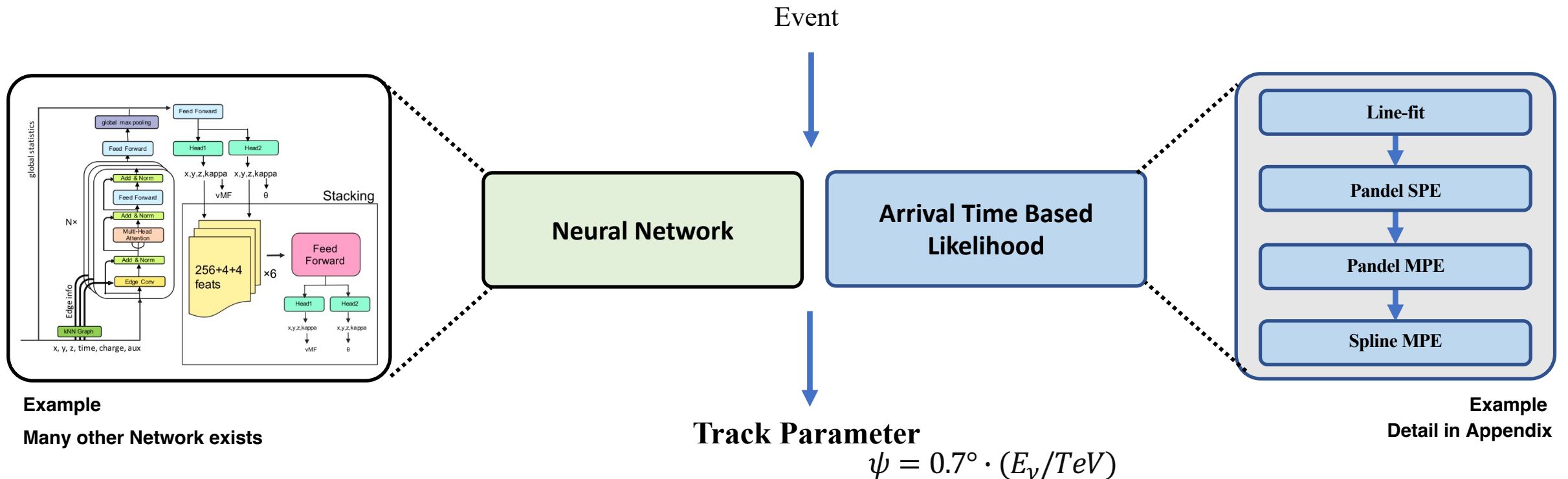


Track Reconstruction in IceCube

- Track events has a angular resolution of $< 1^\circ$ → Main channel for neutrino's source study
- IceCube has mainly two reconstruction methodology for muon reconstruction

1. Likelihood Reconstruction based on Photon Arrival time Modeling

2. Regression Model using Neural Network



Training Objective

- Original objective of FM is

$$L_{FM} = E_{\{t, x_0\}} \|u_{\theta}(x_t, t) - u_t(x_t)\|^2$$

however, where $x_1 \sim q_{data}$ define

$$L_{CFM} = E_{\{t, x_0, x_1\}} \|u_{\theta}(x_t, t) - u_t(x_t | x_1)\|^2$$

then

$$\nabla_{\theta} L_{FM} = \nabla_{\theta} L_{CFM}$$

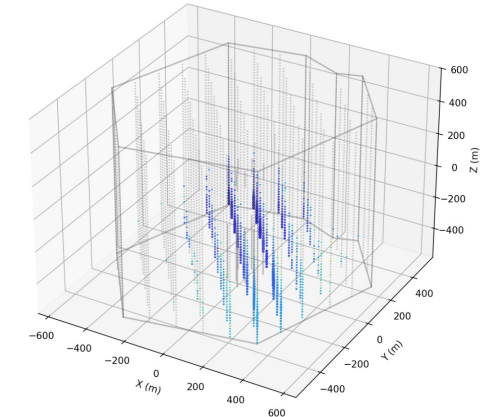
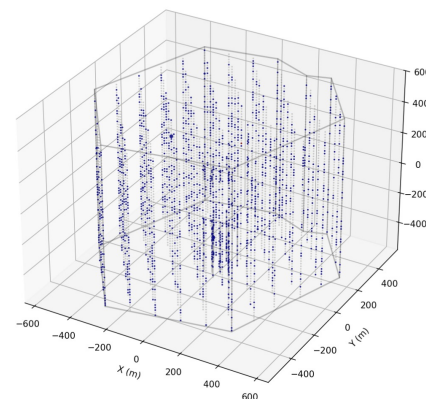
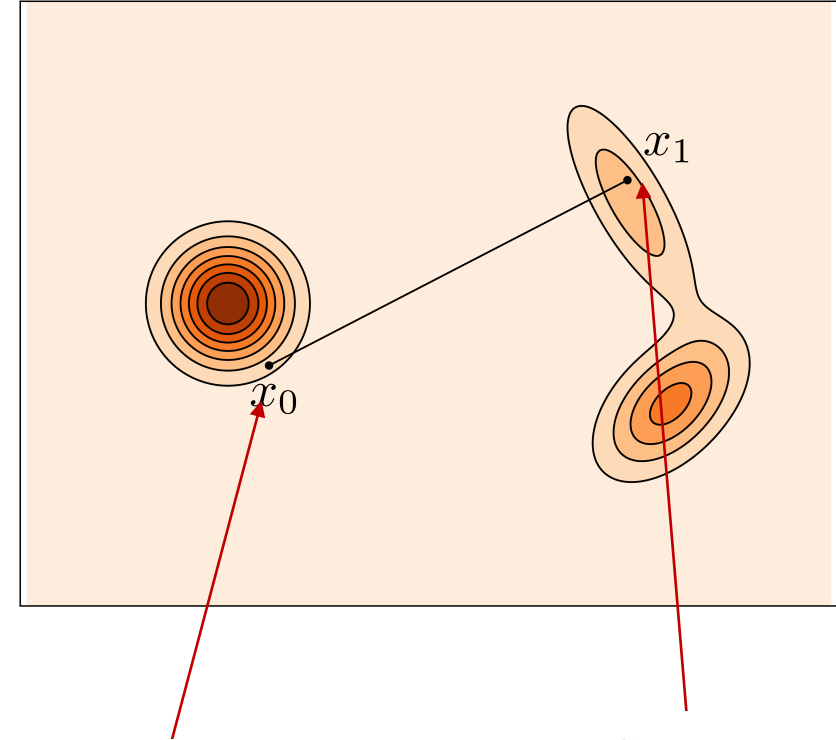
- Let $p_t(x_t | x_1) = N(tx_1, (1-t)^2 I)$

$$x_t = tx_1 + (1-t)x_0 \sim tx_1 + (1-t)N(0, I)$$

$$\dot{x}_t = x_1 - x_0 = u_t(x_t | x_1)$$

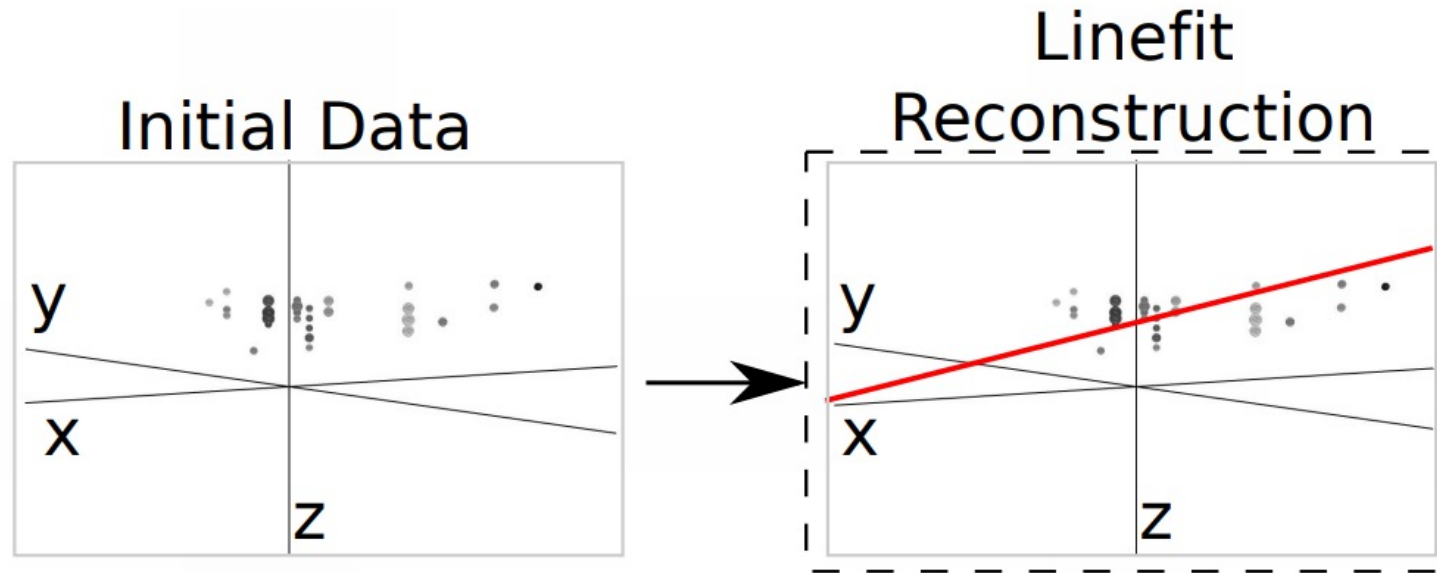
- In conclusion our loss function is in this form

$$L_{CFM} = E_{\{t, x_0, x_1\}} \|u_{\theta}(x_t, t) - (x_1 - x_0)\|^2$$



Line-fit

- o Very first step is called line-fit
- o Simply finding the track that minimizes the squared distance errors between the triggered DOMs



- o Obviously optical ice properties are not taken into account

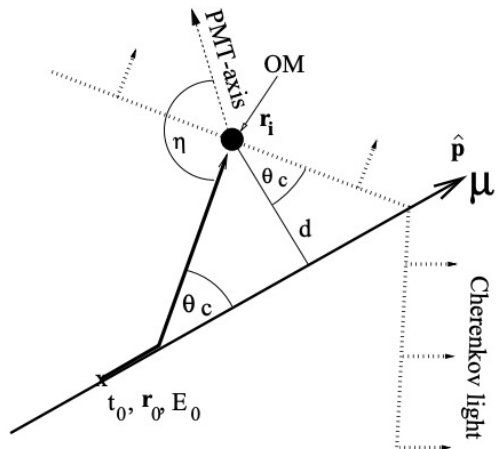
PandelSPE

- o SPE : Single Photo Electron
- o PandelSPE uses the arrival time information of only first Cherenkov photon.

$$\mathcal{L}_{\text{SPE}}(\vec{x}|\vec{\theta}) = \prod_i^{\text{1st hits}} p(t_{\text{res},i}|\vec{\theta}) \quad t_{\text{res}} = t_{\text{hit}} - t_{\text{geo}}$$

Pandel Function

- o Photon arrival time depends on the ice property and relative position between DOM and muon track.
- o Studies have motivated the use of the Pandel function to model the photon arrival time



$$t_{\text{geo}} = t_0 + \frac{\hat{\mathbf{p}} \cdot (\mathbf{r}_i - \mathbf{r}_0) + d \tan \theta_c}{c_{\text{vac}}}$$

$$p(t_{\text{res}}|\vec{\theta}) = \frac{1}{N(d)} \frac{\tau^{-(d/\lambda)} \cdot t_{\text{res}}^{(d/\lambda-1)}}{\Gamma(d/\lambda)} \cdot \exp\left(-t_{\text{res}} \cdot \left(\frac{1}{\tau} + \frac{c_{\text{medium}}}{\lambda_a}\right) - \frac{d}{\lambda_a}\right),$$

$$N(d) = e^{-d/\lambda_a} \cdot \left(1 + \frac{\tau \cdot c_{\text{medium}}}{\lambda_a}\right)^{-d/\lambda},$$

Scattering length, Absorption length, is fixed to the values following

$$\lambda = 33.3 \text{ m} \quad \lambda_a = 98 \text{ m} \quad \tau = 557 \text{ ns}.$$

Only free parameter is d , distance between the track and DOM.

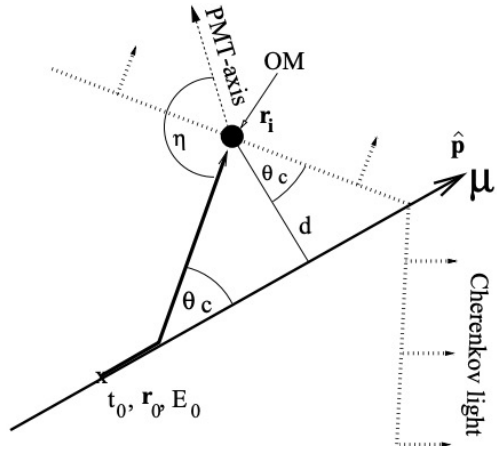
PandelMPE

- o MPE : Single Photo Electron
- o PandelMPE uses the first arrival time information and number of measured photon

$$\mathcal{L}_{\text{MPE}}(\vec{x}|\vec{\theta}) = \prod_i^{\text{1st hits}} n_i \cdot p(t_{\text{res},i}|\vec{\theta}) \cdot (1 - P(t_{\text{res},i}|\vec{\theta}))^{n_i-1} \quad P(t_{\text{res}}|\vec{\theta}) = \int_{-\infty}^{t_{\text{res}}} p(t|\vec{\theta}) dt$$

Pandel Function

- o Photon arrival time depends on the ice property and relative position between DOM and muon track.
- o Studies have motivated the use of the Pandel function to model the photon arrival time



$$t_{\text{geo}} = t_0 + \frac{\hat{\mathbf{p}} \cdot (\mathbf{r}_i - \mathbf{r}_0) + d \tan \theta_c}{c_{\text{vac}}}$$

$$p(t_{\text{res}}|\vec{\theta}) = \frac{1}{N(d)} \frac{\tau^{-(d/\lambda)} \cdot t_{\text{res}}^{(d/\lambda-1)}}{\Gamma(d/\lambda)} \cdot \exp\left(-t_{\text{res}} \cdot \left(\frac{1}{\tau} + \frac{c_{\text{medium}}}{\lambda_a}\right) - \frac{d}{\lambda_a}\right),$$

$$N(d) = e^{-d/\lambda_a} \cdot \left(1 + \frac{\tau \cdot c_{\text{medium}}}{\lambda_a}\right)^{-d/\lambda},$$

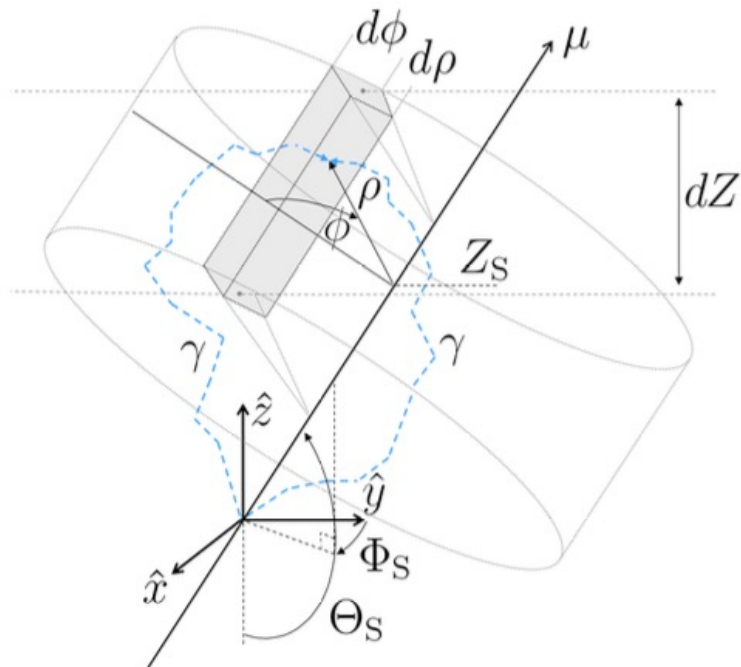
Scattering length, Absorption length, is fixed to the values following

$$\lambda = 33.3 \text{ m} \quad \lambda_a = 98 \text{ m} \quad \tau = 557 \text{ ns}.$$

Only free parameter is d , distance between the track and DOM.

SplineMPE

- o But we know that scattering & absorption length varies over depth.
- o So setting absorption & scattering length fixed , which is done in Pandel MPE, is not great option



- o In this context SplineMPE accrued
SplineMPE uses likelihood formula same as PandelMPE

$$\mathcal{L}_{\text{MPE}}(\vec{x}|\vec{\theta}) = \prod_i^{\text{1st hits}} n_i \cdot p(t_{\text{res},i}|\vec{\theta}) \cdot (1 - P(t_{\text{res},i}|\vec{\theta}))^{n_i-1}$$

But Instead of using Pandel function it creates PDF based on photon distribution table binned as Table 3.1, Also it uses splines (B-spline) technique for the interpolations

Dimension	# bins per dimension		Minimum	Maximum
	old Photonics	this work		
Zenith angle Θ_S	18	18	0°	180°
Depth Z_S	75	75	-850 m	650 m
Azimuthal ϕ	9	36	0°	180°
Distance ρ	30	200	0 m	580 m
Time residual t_{res}	50	210	0 ns	7000 ns

Table 3.1: Photon table binning